

AQIVision

PM_{2.5} Estimation Using Multi-Source Data Fusion Through
Machine Learning

Project Report

Submitted by

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Abstract

Air pollution is one of the most significant environmental challenges affecting public health worldwide. Fine particulate matter ($\text{PM}_{2.5}$) is particularly hazardous due to its ability to penetrate deep into the respiratory system and bloodstream. Accurate estimation of $\text{PM}_{2.5}$ concentration is therefore essential for environmental monitoring and air quality management.

This project presents AQIVision, a machine learning-based framework developed for estimating $\text{PM}_{2.5}$ concentrations using multiple environmental data sources. Ground-based pollutant measurements obtained from the Central Pollution Control Board (CPCB) are combined with satellite-derived Aerosol Optical Depth (AOD) information from the MODIS MAIAC dataset and meteorological variables extracted from NASA's MERRA-2 reanalysis dataset.

The project includes data collection, preprocessing, feature engineering, missing value treatment, satellite data extraction, meteorological data integration, dataset fusion, and Random Forest model development. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

The integration of ground observations, satellite measurements, and meteorological information provides a comprehensive framework for $\text{PM}_{2.5}$ estimation and demonstrates the effectiveness of multi-source data fusion for air quality prediction.

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1. Objective

The primary objective of this project is to develop a machine learning-based framework for estimating monthly $\text{PM}_{2.5}$ concentrations across major Indian cities using multi-source environmental datasets.

The specific objectives are listed below.

- Collect ground-based air quality measurements from the Central Pollution Control Board (CPCB).
- Perform data cleaning and preprocessing to prepare the dataset for machine learning.
- Generate additional temporal and categorical features through feature engineering.
- Handle missing values using appropriate statistical techniques.
- Integrate satellite-derived Aerosol Optical Depth (AOD) obtained from the MODIS MAIAC product.
- Extract meteorological variables from the NASA MERRA-2 reanalysis dataset using Google Earth Engine.
- Merge ground observations, satellite measurements, and meteorological parameters into a unified dataset.
- Develop a Random Forest Regression model for $\text{PM}_{2.5}$ estimation.
- Evaluate model performance using MAE, RMSE, and R^2 metrics.
- Identify the most influential environmental factors affecting $\text{PM}_{2.5}$ concentration through feature importance analysis.

2. Project Overview

AQIVision is a machine learning-based air quality estimation framework designed to predict $\text{PM}_{2.5}$ concentrations by integrating heterogeneous environmental datasets.

The workflow begins with collecting air quality observations from CPCB monitoring stations. The collected data undergo cleaning, preprocessing, feature engineering, and missing value treatment. Satellite-derived Aerosol Optical Depth values are extracted from the MODIS MAIAC dataset using Google Earth Engine, while meteorological variables are obtained from the NASA MERRA-2 reanalysis dataset.

After integrating these datasets, a Random Forest Regression model is trained to estimate $\text{PM}_{2.5}$ concentrations. The developed model is evaluated using multiple statistical performance metrics and interpreted using feature importance analysis.

3. Data Collection

The primary source of air quality information used in this study is the Central Pollution Control Board (CPCB) of India.

The following datasets were obtained.

Table 3.1: Datasets Used

Dataset	Description
city_day.csv	Daily city-wise air quality data
station_day.csv	Daily station-wise air quality data
station_hour.csv	Hourly station-wise air quality data
city_hour.csv	Hourly city-wise air quality data
stations.csv	Station information and metadata

Among these datasets, the city_day.csv dataset was selected as the primary source for developing the PM_{2.5} estimation model because it provides city-level daily pollutant measurements required for subsequent preprocessing and machine learning analysis.

3.1 Data Cleaning

Data cleaning was performed to improve the quality and consistency of the dataset before machine learning model development.

The following preprocessing operations were carried out.

- Latitude and longitude values were added for every city.
- Records with missing PM_{2.5} observations were removed.
- Data consistency was verified.
- Duplicate records were checked.
- Dataset dimensions were validated after cleaning.

Removing records with unavailable PM_{2.5} values ensured that every observation contained a valid target variable required for supervised learning.

4. Data Preprocessing

Several preprocessing techniques were applied before model development.

4.1 Feature Engineering

The Date column was converted into datetime format, and three additional temporal variables were generated.

- Year
- Month
- Day

These features enable the model to capture long-term trends, seasonal variations, and daily changes in PM_{2.5} concentration.

4.2 Categorical Encoding

The City attribute is categorical in nature and therefore cannot be directly used by machine learning algorithms.

Label Encoding was applied to convert city names into numerical identifiers, resulting in a new feature named

City_Encoded.

This encoded feature preserves city-specific information while making the dataset compatible with Random Forest Regression.

4.3 Missing Value Analysis

Missing values are a common challenge in environmental monitoring datasets and can significantly affect the performance of machine learning models if left untreated. Before training the Random Forest model, a detailed analysis was conducted to quantify the extent of missing data in each pollutant variable.

The objective of this analysis was to determine the completeness of the dataset, identify variables with a high proportion of missing observations, and decide whether these variables could be retained for subsequent model development after appropriate imputation.

The following pollutant variables were analysed:

- PM₁₀
- NO
- NO₂
- NO_x
- NH₃
- CO
- SO₂
- O₃

The number and percentage of missing values for each pollutant feature are summarized in Table 4.1.

Table 4.1: Missing Value Statistics for Pollutant Variables

Feature	Missing Values	Missing Percentage
CO	2059	6.97%
NO	3582	12.13%
NO ₂	3585	12.14%
SO ₂	3854	13.05%
O ₃	4022	13.62%
NO _x	4185	14.17%
NH ₃	10328	34.97%
PM ₁₀	11140	37.72%

The analysis produced the following observations:

- CO contained the lowest percentage of missing observations (6.97%), indicating relatively complete measurements.
- NO, NO₂, SO₂, O₃, and NO_x exhibited moderate levels of missing data ranging from approximately 12% to 14%.
- NH₃ and PM₁₀ contained comparatively higher proportions of missing observations.

- None of the selected pollutant variables exceeded 50% missing data.
- Therefore, all pollutant variables were retained and considered suitable for subsequent imputation and machine learning analysis.

4.4 Missing Value Visualization

To obtain a clearer understanding of the completeness of each pollutant variable, a graphical visualization of the missing values was generated. A bar chart was used to compare the number of missing observations across all selected pollution features.

Figure 4.1 illustrates the distribution of missing values.

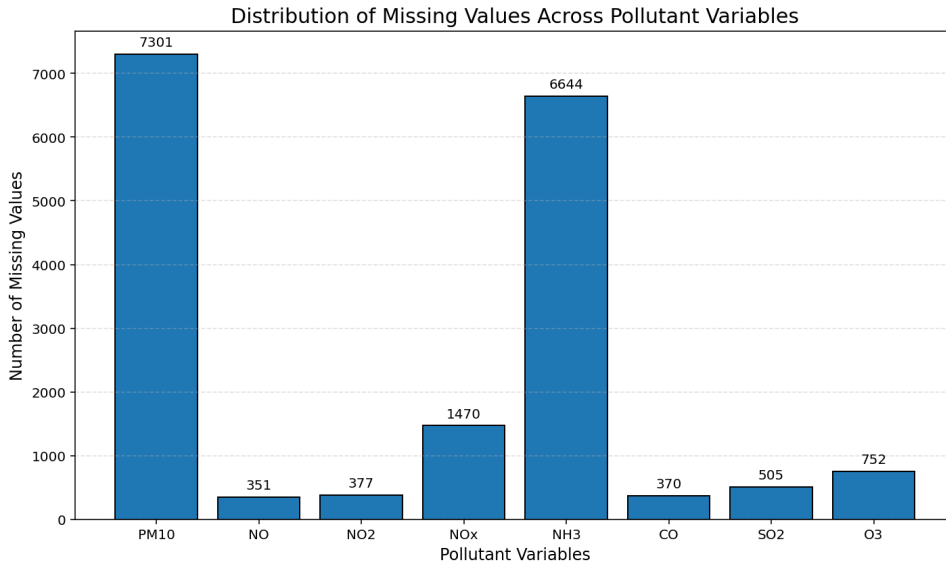


Figure 4.1: Distribution of Missing Values Across Pollutant Variables

The visualization reveals that PM₁₀ contains the largest number of missing observations, followed by NH₃. Carbon Monoxide (CO) exhibits the highest data completeness among all variables, while the remaining pollutant features show comparable levels of missing data.

Overall, the visualization confirms that the available observations are sufficient for reliable statistical imputation and subsequent machine learning model development.

4.5 Missing Value Treatment

Following the missing value analysis, an appropriate strategy was required to handle incomplete observations while preserving the maximum amount of information contained in the dataset.

Median imputation was selected because air pollutant concentrations often exhibit skewed distributions and extreme values resulting from episodic pollution events. Unlike the arithmetic mean, the median is less sensitive to outliers and therefore provides a more robust estimate of the central tendency.

The advantages of median imputation include:

- Robustness against extreme pollution values.
- Preservation of the complete dataset.
- Reduction of information loss.
- Improved stability during Random Forest training.

After applying median imputation, all selected pollutant variables contained zero missing values and the dataset became fully prepared for machine learning model development.

4.6 Random Forest Model Development

Following data preprocessing and missing value treatment, a Random Forest Regression model was developed to estimate $PM_{2.5}$ concentrations. Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees to improve prediction accuracy while reducing the risk of overfitting.

The algorithm was selected because of its ability to model complex nonlinear relationships, handle high-dimensional datasets, and provide reliable estimates of feature importance.

4.6.1 Input Features

The model was trained using twelve predictor variables obtained after preprocessing and feature engineering.

Table 4.2: Predictor Variables Used for Random Forest Model

Feature	Description
City_Encoded	Encoded city identifier
Year	Year of observation
Month	Month of observation
Day	Day of observation
PM ₁₀	Particulate Matter (10 μm)
NO	Nitric Oxide
NO ₂	Nitrogen Dioxide
NO _x	Nitrogen Oxides
NH ₃	Ammonia
CO	Carbon Monoxide
SO ₂	Sulphur Dioxide
O ₃	Ozone

The target variable used for model training was PM_{2.5} concentration.

4.6.2 Model Training

The dataset was divided into training and testing subsets using an 80:20 split. The training subset was used to construct the Random Forest model, while the testing subset was reserved for evaluating its predictive performance on unseen data.

The Random Forest model consisted of 200 decision trees. During training, each tree was generated using a bootstrap sample of the training data, and random feature selection was performed at every node split. The final prediction was obtained by averaging the outputs of all decision trees.

4.6.3 Model Evaluation Metrics

The predictive performance of the model was evaluated using three standard regression metrics.

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination (R^2)

Mean Absolute Error measures the average magnitude of prediction errors, while Root Mean Square Error provides greater emphasis on larger prediction errors. The coefficient of determination (R^2) measures the proportion of variance in PM_{2.5} concentrations explained by the model.

4.7 Baseline Random Forest Results

After incorporating eight additional pollutant variables and performing median imputation, the Random Forest model demonstrated improved predictive performance compared with the baseline model.

The obtained performance metrics are presented in Table 4.3.

Table 4.3: Performance of the Improved Random Forest Model

Metric	Value
Mean Absolute Error (MAE)	13.86
Root Mean Square Error (RMSE)	31.89
Coefficient of Determination (R^2)	0.763

The results indicate a substantial improvement in model performance after incorporating additional pollutant variables.

The model successfully explains approximately 76.3% of the variability in $\text{PM}_{2.5}$ concentrations across the study area. Furthermore, reductions in both MAE and RMSE demonstrate improved predictive accuracy and lower estimation error.

These findings confirm that incorporating additional atmospheric pollutant measurements significantly enhances the capability of the Random Forest model to estimate $\text{PM}_{2.5}$ concentrations.

4.8 Feature Importance Analysis

One of the major advantages of the Random Forest algorithm is its ability to estimate the relative importance of each predictor variable. Feature importance analysis helps identify the variables that contribute most significantly to the prediction of $\text{PM}_{2.5}$ concentrations.

After training the Random Forest model, the importance score of each predictor variable was computed. These scores indicate the contribution of each feature toward reducing prediction error during the tree-building process.

The feature importance values obtained from the trained model are presented in Table 4.4.

Table 4.4: Feature Importance Scores of the Random Forest Model

Rank	Feature	Importance
1	PM ₁₀	0.451
2	NO	0.093
3	City_Encoded	0.082
4	Month	0.080
5	CO	0.065
6	NO ₂	0.051
7	O ₃	0.039
8	SO ₂	0.038
9	NO _x	0.032
10	Day	0.029
11	NH ₃	0.023
12	Year	0.017

For better visualization, the feature importance scores are illustrated in Figure 4.2.

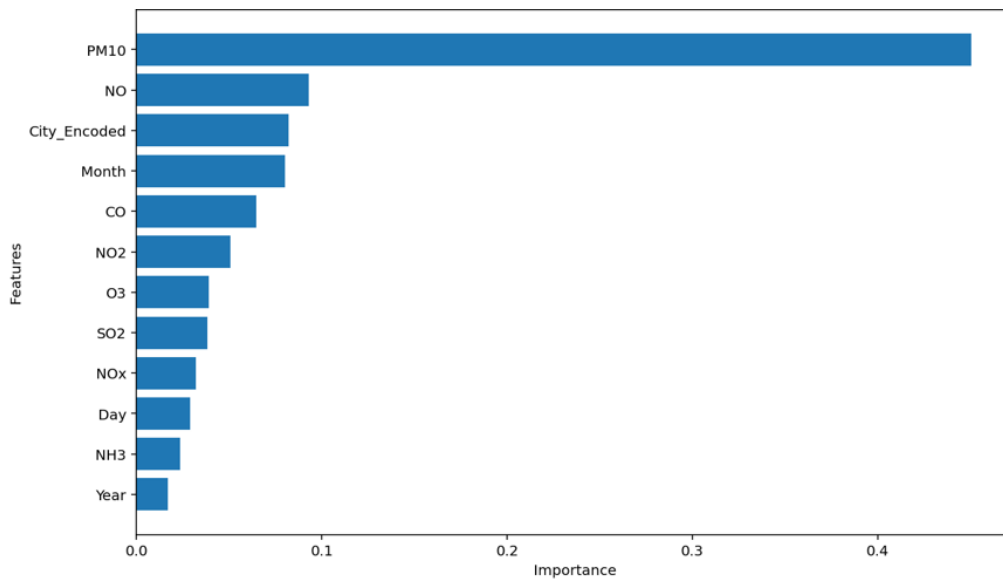


Figure 4.2: Feature Importance Obtained from the Random Forest Model

4.8.1 Discussion

The feature importance analysis provides valuable insights into the factors influencing PM_{2.5} concentration.

PM₁₀ emerged as the most influential predictor with an importance score of 0.451, accounting for nearly half of the model's predictive capability. This observation is scientifically reasonable because PM₁₀ and PM_{2.5} originate from many common emission sources, including vehicular traffic, industrial activities, road dust, and construction operations.

Nitric Oxide (NO) was identified as the second most important feature. Since NO is

primarily emitted from vehicle exhaust and fuel combustion, its high importance indicates that traffic-related emissions play a significant role in determining $\text{PM}_{2.5}$ levels.

The encoded city identifier also contributed substantially to the prediction process, suggesting that geographical location influences air quality due to differences in industrial activity, traffic density, land use, and local environmental conditions.

Seasonal variation was captured through the Month feature, demonstrating that $\text{PM}_{2.5}$ concentrations vary considerably throughout the year because of changing meteorological conditions such as winter pollution accumulation, monsoon washout, and enhanced atmospheric mixing during summer.

The Year feature exhibited the lowest importance score, indicating that long-term temporal variation was relatively small compared with pollutant concentrations and seasonal effects within the study period.

Overall, the feature importance analysis confirms that particulate pollutants, traffic emissions, geographical location, and seasonal variability are the dominant factors influencing $\text{PM}_{2.5}$ concentration in the developed Random Forest model.

4.9 Actual vs Predicted Scatter Plot

To evaluate the predictive capability of the Random Forest model, the predicted $\text{PM}_{2.5}$ concentrations were compared with the corresponding observed values using a scatter plot.

Figure 4.3 presents the relationship between actual and predicted $\text{PM}_{2.5}$ concentrations.

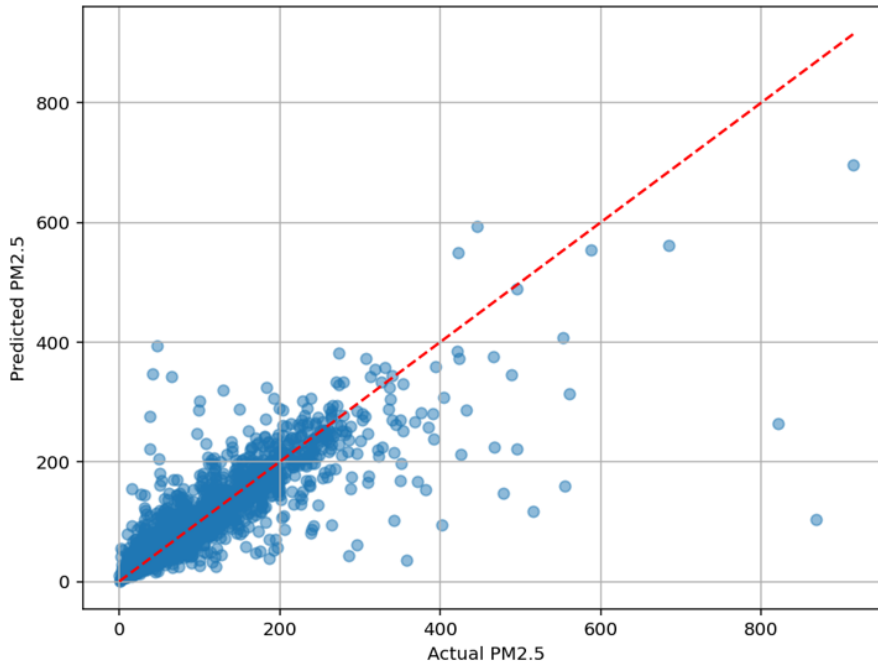


Figure 4.3: Comparison of Actual and Predicted $\text{PM}_{2.5}$ Concentrations

The red dashed line represents the ideal condition where predicted values are exactly equal to the observed values.

Most of the data points are concentrated close to the reference line, indicating that the Random Forest model is capable of accurately estimating $\text{PM}_{2.5}$ concentrations for the majority of observations.

A small number of points deviate from the ideal line, particularly at higher $\text{PM}_{2.5}$ concentrations. Such deviations are expected because extreme pollution events occur less frequently and are inherently more difficult for machine learning models to predict.

Overall, the scatter plot demonstrates that the developed Random Forest model successfully captures the overall trend of $\text{PM}_{2.5}$ concentrations and provides reliable predictive performance across the study dataset.

5. MODIS MAIAC Aerosol Optical Depth (AOD) Integration

5.1 Introduction

Ground-based air quality monitoring stations provide accurate measurements of $\text{PM}_{2.5}$ concentrations; however, their spatial coverage is limited to specific monitoring locations. To improve the spatial representation of atmospheric aerosols and enhance the predictive capability of the Random Forest model, satellite-derived Aerosol Optical Depth (AOD) data were incorporated into the AQIVision framework.

Aerosol Optical Depth represents the degree to which aerosols attenuate incoming solar radiation while passing through the atmosphere. Higher AOD values generally indicate greater concentrations of suspended particulate matter and therefore provide valuable information for estimating $\text{PM}_{2.5}$ concentrations.

The MODIS MAIAC aerosol product was selected because of its high spatial resolution, improved atmospheric correction algorithm, and widespread use in air quality studies. Satellite data were accessed and processed using Google Earth Engine, enabling efficient extraction of monthly AOD values for all study locations.

5.2 Google Earth Engine Platform

Google Earth Engine (GEE) is a cloud-based geospatial processing platform that provides direct access to large collections of satellite imagery and environmental datasets. The platform eliminates the need to download large volumes of remote sensing data and provides powerful tools for large-scale geospatial analysis.

In this project, Google Earth Engine was used to:

- Access the MODIS MAIAC satellite archive.
- Filter imagery according to the study period.
- Extract monthly Aerosol Optical Depth values.

- Perform city-wise spatial analysis.
- Export the extracted data for subsequent machine learning applications.

The use of Google Earth Engine significantly reduced computational complexity and enabled automated extraction of satellite observations for all selected cities.

5.3 Selection of MODIS MAIAC Dataset

Several remotely sensed aerosol datasets are available for atmospheric studies. Among these, the MODIS MAIAC product was selected because it provides high-quality aerosol retrievals with improved spatial resolution compared to conventional MODIS aerosol products.

Table 5.1 summarizes the selected dataset.

Table 5.1: MODIS MAIAC Dataset Used in the Study

Dataset Name	MODIS/061/MCD19A2_GRANULES
Satellite	Terra and Aqua MODIS
Data Provider	NASA
Processing Platform	Google Earth Engine
Spatial Resolution	1 km
Variable Used	Optical_Depth_055
Study Period	January 2015 – June 2020

The MAIAC algorithm performs advanced atmospheric correction and aerosol retrieval, making it highly suitable for estimating particulate matter concentrations over urban environments.

5.4 Identification of AOD Variable

The selected MODIS MAIAC dataset contains multiple spectral and quality assessment bands. To determine the variable appropriate for $PM_{2.5}$ estimation, all available bands were examined within Google Earth Engine.

Among the available variables, the **Optical_Depth_055** band was selected for further analysis.

This variable represents Aerosol Optical Depth measured at a wavelength of 550 nm, which is widely accepted as a standard indicator of atmospheric aerosol loading. Since atmospheric aerosols are closely associated with particulate matter concentrations, the Optical_Depth_055 band provides an important predictor for $PM_{2.5}$ estimation.

The selected variable was subsequently renamed as **AOD_550** within the AQIVision dataset for consistency during machine learning model development.

5.5 Preparation of AQIVision City Coordinates

Satellite data extraction requires accurate geographic coordinates corresponding to each study location. Therefore, latitude and longitude values were compiled for all selected AQIVision cities.

A comma-separated values (CSV) file named **aqivision_cities.csv** was created containing the following attributes:

- City Name
- Latitude
- Longitude

These coordinates served as the spatial reference locations for extracting monthly Aerosol Optical Depth values from the MODIS MAIAC dataset.

Preparing a standardized coordinate file ensured consistent extraction of satellite observations across all cities included in the study.

5.6 Uploading City Coordinates to Google Earth Engine

The prepared CSV file containing city coordinates was uploaded to Google Earth Engine as a Feature Collection.

During the upload process, the longitude column was assigned as the X-coordinate, while the latitude column was assigned as the Y-coordinate using the EPSG:4326 geographic coordinate reference system.

After successful upload, the asset was stored within the Google Earth Engine project as:

```
projects/aqivision/assets/aqivision_cities
```

The uploaded Feature Collection contained twenty-six city locations, each represented by a point geometry and associated city name.

5.7 Verification of AOD Extraction

Before performing large-scale extraction, the satellite retrieval procedure was validated using a single monthly observation.

A sample city was selected to verify that the MODIS MAIAC dataset, selected AOD variable, and uploaded city coordinates were correctly linked.

Successful extraction confirmed that:

- the selected dataset was accessible,
- the Optical_Depth_055 band was correctly identified,
- city coordinates were accurately referenced,
- monthly filtering operated correctly, and
- Aerosol Optical Depth values could be extracted without errors.

After successful verification, the same methodology was applied to all AQIVision cities.

5.8 Monthly AOD Extraction

Monthly Aerosol Optical Depth values were extracted for all twenty-six AQIVision cities over the study period extending from January 2015 to June 2020.

For each month, satellite observations were filtered according to the selected temporal interval, and the mean Aerosol Optical Depth was computed for every city location.

The extracted dataset contained the following attributes:

- City
- Month
- AOD_550

The resulting dataset provides monthly satellite-derived aerosol information corresponding to each city included in the study.

5.9 Integration of AOD with the PM Dataset

Following successful extraction, the monthly Aerosol Optical Depth dataset was merged with the monthly PM_{2.5} dataset using the common attributes **City** and **Month**.

A left join strategy was adopted to preserve all PM_{2.5} observations while appending the corresponding AOD values wherever available.

The merged dataset consisted of ground-based pollutant measurements together with satellite-derived aerosol information, providing a richer representation of atmospheric conditions for machine learning model development.

The resulting integrated dataset served as the foundation for the subsequent incorporation of meteorological variables from the NASA MERRA-2 reanalysis dataset.

6. MERRA-2 Meteorological Data Integration

6.1 Introduction

Although ground-based pollutant measurements and satellite-derived Aerosol Optical Depth (AOD) provide valuable information for estimating $\text{PM}_{2.5}$ concentrations, atmospheric conditions also play a critical role in determining the transport, dispersion, formation, and accumulation of air pollutants. Meteorological parameters influence the physical and chemical processes governing particulate matter and therefore serve as important predictors in air quality modeling.

To capture these atmospheric influences, meteorological variables from the Modern-Era Retrospective Analysis for Research and Applications Version 2 (MERRA-2) dataset were incorporated into the AQIVision framework. The extracted meteorological variables were integrated with the existing $\text{PM}_{2.5}$ and satellite-derived AOD datasets to develop a comprehensive multi-source dataset for machine learning.

6.2 Selection of the MERRA-2 Dataset

NASA’s MERRA-2 reanalysis dataset was selected because it provides globally consistent atmospheric observations by combining satellite measurements, numerical weather prediction models, and ground-based observations.

The dataset is available through Google Earth Engine and contains a wide range of meteorological variables suitable for environmental and air quality studies.

The characteristics of the selected dataset are summarized in Table [6.1](#).

Table 6.1: MERRA-2 Dataset Used in this Study

Dataset	NASA/GSFC/MERRA/slv/2
Data Provider	NASA Global Modeling and Assimilation Office
Platform	Google Earth Engine
Dataset Type	Surface Layer Diagnostics
Study Period	January 2015 – June 2020
Coverage	Global
Temporal Resolution	Hourly

The MERRA-2 Surface Layer Diagnostics product was selected because it contains meteorological variables that directly influence $PM_{2.5}$ formation and atmospheric dispersion.

6.3 Selection of Meteorological Variables

Initially, all available meteorological bands within the selected MERRA-2 dataset were explored. Approximately forty-seven atmospheric variables were available for analysis.

Among these variables, six meteorological parameters were selected based on their relevance to $PM_{2.5}$ estimation and their influence on atmospheric processes.

Table 6.2: Selected Meteorological Variables

Variable	Description
T2M	Air Temperature at 2 m
QV2M	Specific Humidity at 2 m
PS	Surface Pressure
PBLTOP	Planetary Boundary Layer Height
U10M	East-West Wind Component
V10M	North-South Wind Component

These variables were selected because they directly influence atmospheric stability, aerosol transport, pollutant dispersion, chemical reactions, and the accumulation of particulate matter near the Earth’s surface.

6.4 Pilot Extraction of Meteorological Data

Before extracting meteorological information for the complete study area, the extraction procedure was validated using a single city. Ahmedabad was selected as the pilot location for verifying the extraction methodology.

Monthly mean meteorological values were extracted for January 2015 using the selected MERRA-2 variables.

Successful extraction confirmed that the selected variables, temporal filtering procedure, and city coordinate references were functioning correctly within Google Earth Engine.

6.5 Interpretation of Selected Meteorological Variables

Each selected meteorological parameter contributes differently to the prediction of $\text{PM}_{2.5}$ concentrations.

Air Temperature (T2M)

Air temperature affects atmospheric mixing, chemical reaction rates, and the formation of secondary aerosols. Temperature variations influence the dispersion and transformation of atmospheric pollutants.

Specific Humidity (QV2M)

Specific humidity represents the moisture content of the atmosphere. Increased humidity promotes hygroscopic growth of aerosol particles, often resulting in elevated $\text{PM}_{2.5}$ concentrations.

Surface Pressure (PS)

Surface pressure provides information regarding atmospheric stability. Stable high-pressure systems generally suppress vertical mixing, allowing pollutants to accumulate near the ground.

Planetary Boundary Layer Height (PBLTOP)

The planetary boundary layer determines the vertical extent within which pollutants are mixed. Lower boundary layer heights generally lead to higher pollutant concentrations because atmospheric mixing is restricted.

Wind Components (U10M and V10M)

The horizontal wind components describe wind movement in east-west and north-south directions. Wind plays an important role in transporting pollutants and reducing local pollutant accumulation through atmospheric dispersion.

6.6 Full-Scale Meteorological Data Extraction

After successful validation, the extraction methodology was extended to all AQIVision cities.

Monthly meteorological observations were extracted for twenty-six cities covering the period from January 2015 to June 2020.

The extraction workflow consisted of the following steps:

1. Loading the AQIVision city locations.
2. Filtering the MERRA-2 dataset according to the required month.
3. Computing monthly mean meteorological values.
4. Extracting meteorological parameters for each city.
5. Repeating the procedure for every month throughout the study period.
6. Exporting the generated dataset for integration with the machine learning framework.

Automated extraction within Google Earth Engine enabled efficient processing of all city-month combinations while maintaining consistency throughout the study period.

6.7 Generated Meteorological Dataset

The completed extraction process generated a comprehensive monthly meteorological dataset containing observations for all study locations.

The dataset characteristics are summarized in Table 6.3.

Table 6.3: Summary of the Extracted Meteorological Dataset

Study Period	January 2015 – June 2020
Number of Cities	26
Number of Months	66
Total Records	1716
Meteorological Variables	6

The generated dataset contains the following attributes:

- City
- Month
- T2M

- QV2M
- PS
- PBLTOP
- U10M
- V10M

The extracted meteorological dataset provides comprehensive atmospheric information describing temperature, humidity, pressure, boundary layer characteristics, and wind conditions for each city-month observation.

This dataset forms the final environmental component required for developing a multi-source $\text{PM}_{2.5}$ estimation framework and is subsequently integrated with the CPCB pollutant dataset and the MODIS MAIAC Aerosol Optical Depth dataset.

7. Multi-Source Dataset Integration

7.1 Introduction

The prediction of $\text{PM}_{2.5}$ concentrations depends on several environmental factors rather than a single source of information. Ground-based pollutant measurements provide direct observations of air quality, satellite-derived Aerosol Optical Depth (AOD) describes atmospheric aerosol loading, and meteorological variables explain atmospheric conditions affecting pollutant transport and dispersion.

To exploit the complementary strengths of these datasets, a multi-source data fusion framework was developed. The integration process combined ground observations obtained from the Central Pollution Control Board (CPCB), MODIS MAIAC Aerosol Optical Depth measurements, and NASA MERRA-2 meteorological parameters into a unified dataset suitable for machine learning.

7.2 Integration of PM Dataset with MODIS MAIAC AOD

The first stage of data fusion involved integrating the monthly $\text{PM}_{2.5}$ dataset with the monthly MODIS MAIAC Aerosol Optical Depth dataset.

Since both datasets were available at monthly temporal resolution, they were merged using two common attributes:

- City
- Month

A left join operation was performed to preserve all observations contained in the monthly pollutant dataset while appending the corresponding AOD values wherever available.

Table [7.1](#) summarizes the first integration stage.

Table 7.1: Integration of PM Dataset with MODIS MAIAC AOD

Input Dataset 1	monthly_pm.xlsx
Input Dataset 2	AQIVision_Monthly_AOD_2015_2020.csv
Merge Keys	City, Month
Join Type	Left Join
Output Dataset	AQIVision_PM_AOD_Merged.xlsx
Total Records	893

The resulting dataset combined ground-based pollutant measurements with satellite-derived aerosol information, providing an improved representation of atmospheric particulate loading.

7.3 Integration with MERRA-2 Meteorological Dataset

Following the successful incorporation of satellite observations, the merged PM–AOD dataset was further integrated with meteorological parameters extracted from the NASA MERRA-2 reanalysis dataset.

The integration again employed the common attributes City and Month to ensure correct temporal and spatial correspondence between observations.

A left join operation was performed so that all records from the PM–AOD dataset were retained while matching meteorological observations were appended.

Table 7.2 summarizes the second stage of data integration.

Table 7.2: Integration of PM–AOD Dataset with MERRA-2 Meteorological Data

Input Dataset 1	AQIVision_PM_AOD_Merged.xlsx
Input Dataset 2	AQIVision_MERRA2_2015_2020.csv
Merge Keys	City, Month
Join Type	Left Join
Output Dataset	AQIVision_Final_Dataset.xlsx
Total Records	893

This integration produced the complete environmental dataset used for the subsequent stages of machine learning model development.

7.4 Final Integrated Dataset

The final AQIVision dataset combines information from three independent environmental data sources into a single machine learning dataset.

The integrated dataset contains:

- Ground-based pollutant measurements obtained from CPCB.
- Satellite-derived Aerosol Optical Depth (AOD₅₅₀) obtained from the MODIS MA-IAC product.
- Meteorological parameters extracted from NASA MERRA-2.

Table 7.3 lists the variables included in the final integrated dataset.

Table 7.3: Variables Included in the Final AQIVision Dataset

Variable	Description
PM _{2.5}	Target Variable
PM ₁₀	Particulate Matter
NO	Nitric Oxide
NO ₂	Nitrogen Dioxide
NO _x	Nitrogen Oxides
NH ₃	Ammonia
CO	Carbon Monoxide
SO ₂	Sulphur Dioxide
O ₃	Ozone
AOD ₅₅₀	Aerosol Optical Depth
T2M	Air Temperature
QV2M	Specific Humidity
PS	Surface Pressure
PBLTOP	Planetary Boundary Layer Height
U10M	East-West Wind Component
V10M	North-South Wind Component

The integrated dataset provides a comprehensive description of air quality conditions by combining pollutant concentrations, atmospheric aerosols, and meteorological factors within a single framework.

7.5 Dataset Characteristics

The characteristics of the final integrated dataset are summarized in Table 7.4.

Table 7.4: Summary of the Final AQIVision Dataset

Study Period	January 2015 – June 2020
Cities	26
Monthly Observations	893
Meteorological Variables	6
Satellite Variable	1
Pollutant Variables	8
Target Variable	PM _{2.5}

The resulting dataset represents one of the primary contributions of the AQIVision project, integrating heterogeneous environmental information into a unified machine learning-ready database.

7.6 Significance of Multi-Source Data Fusion

Ground monitoring stations provide highly accurate pollutant measurements but suffer from limited spatial coverage. Satellite observations offer regional aerosol information but cannot directly measure $\text{PM}_{2.5}$ at ground level. Meteorological parameters describe atmospheric processes responsible for pollutant transport, dispersion, and accumulation.

By integrating these complementary data sources, the AQIVision framework captures multiple environmental factors influencing $\text{PM}_{2.5}$ concentrations.

The principal advantages of the proposed multi-source data fusion approach include:

- Improved representation of atmospheric conditions.
- Inclusion of satellite-derived aerosol information.
- Incorporation of meteorological influences on pollutant transport.
- Enhanced predictive capability of the Random Forest model.
- Creation of a comprehensive machine learning dataset suitable for environmental prediction.

The final integrated dataset forms the basis for developing the final Random Forest model presented in the next chapter.

8. Final Random Forest Model Development

8.1 Introduction

The previous chapters presented the development of an initial Random Forest model using ground-based pollutant measurements, followed by the integration of satellite-derived Aerosol Optical Depth (AOD) data and meteorological parameters extracted from the NASA MERRA-2 reanalysis dataset.

The objective of this chapter is to develop the final Random Forest model using the integrated multi-source dataset. By combining pollutant measurements, satellite observations, and meteorological information, the model is expected to achieve improved prediction accuracy and provide a more comprehensive representation of the environmental factors influencing $\text{PM}_{2.5}$ concentrations.

8.2 Final Dataset Preparation

Following the completion of dataset integration, the final AQIVision dataset was prepared for machine learning model development.

The integrated dataset contains:

- Ground-based pollutant measurements obtained from CPCB.
- Satellite-derived Aerosol Optical Depth (AOD_{550}).
- Meteorological variables extracted from NASA MERRA-2.
- Temporal variables generated through feature engineering.

The dataset was examined to verify data consistency, identify missing observations, and ensure compatibility with the Random Forest Regression algorithm.

8.3 Missing Value Analysis

Following dataset integration, a second missing value analysis was performed to evaluate the completeness of the newly added satellite and meteorological variables.

Unlike the original CPCB pollutant dataset, the integrated dataset contains missing observations resulting primarily from unavailable satellite retrievals and occasional gaps in meteorological observations.

Table 8.1 summarizes the missing values observed in the integrated dataset.

Table 8.1: Missing Values in the Final Integrated Dataset

Variable	Missing Values
AOD ₅₅₀	63
T2M	23
QV2M	23
PS	23
PBLTOP	23
U10M	23
V10M	23
Wind Speed	23

The analysis indicates that only a small proportion of observations contain missing values. Since the percentage of missing data was relatively low, all variables were retained for model development after applying statistical imputation.

8.4 Feature Engineering

Additional feature engineering was performed to enhance the predictive capability of the Random Forest model.

Besides the temporal variables generated during preprocessing, a new meteorological feature representing wind speed was derived from the horizontal wind components.

Wind speed was calculated using the magnitude of the east-west and north-south wind vectors according to Equation 8.1.

$$\text{WindSpeed} = \sqrt{U10M^2 + V10M^2} \quad (8.1)$$

The derived Wind Speed feature provides a direct measure of atmospheric transport intensity and is expected to improve the representation of pollutant dispersion.

The final predictor variables used for machine learning are summarized in Table 8.2.

Table 8.2: Final Predictor Variables

Feature	Description
City_Encoded	Encoded city identifier
Year	Year
Month	Month
PM ₁₀	Particulate Matter
NO	Nitric Oxide
NO ₂	Nitrogen Dioxide
NO _x	Nitrogen Oxides
NH ₃	Ammonia
CO	Carbon Monoxide
SO ₂	Sulphur Dioxide
O ₃	Ozone
AOD ₅₅₀	Aerosol Optical Depth
T2M	Air Temperature
QV2M	Specific Humidity
PS	Surface Pressure
PBLTOP	Boundary Layer Height
Wind Speed	Derived Wind Speed

These variables collectively represent emissions, atmospheric aerosol loading, meteorological conditions, geographical variation, and temporal variability, providing a comprehensive feature set for PM_{2.5} estimation.

8.5 Final Random Forest Model Development

The final Random Forest Regression model was developed using the integrated AQIVision dataset containing pollutant measurements, satellite-derived Aerosol Optical Depth (AOD), and meteorological variables.

Random Forest was selected because of its robustness, ability to model nonlinear relationships, resistance to overfitting, and capability of handling heterogeneous environmental datasets.

Prior to model training, categorical variables were encoded, missing values were treated using median imputation, and all predictor variables were assembled into a single feature matrix.

The integrated dataset was divided into training and testing subsets using an 80:20 ratio. The training subset was used for model development, while the testing subset was reserved for evaluating the predictive capability of the model on previously unseen observations.

The Random Forest model consisted of 200 decision trees with a fixed random seed to ensure reproducibility of the experimental results.

8.6 Performance Evaluation

The predictive performance of the developed Random Forest model was evaluated using three standard regression metrics.

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination (R^2)

The obtained performance metrics are summarized in Table 8.3.

Table 8.3: Performance of the Final Random Forest Model

Metric	Value
Mean Absolute Error (MAE)	13.32
Root Mean Square Error (RMSE)	20.53
Coefficient of Determination (R^2)	0.820

The final Random Forest model explains approximately 82% of the variability in $\text{PM}_{2.5}$ concentration across the study area.

Compared with the initial Random Forest model, reductions in both MAE and RMSE demonstrate improved predictive accuracy, while the increase in the coefficient of determination confirms that incorporating satellite-derived aerosol information and meteorological variables substantially enhances model performance.

8.7 Cross-Validation

To evaluate the robustness and generalization capability of the developed Random Forest model, five-fold cross-validation was performed.

The complete dataset was divided into five approximately equal subsets. During each iteration, four subsets were used for training while the remaining subset was used for validation. This procedure was repeated five times so that each subset served as the validation set exactly once.

Cross-validation minimizes the influence of random train-test splits and provides a more reliable estimate of model performance.

The average cross-validation score demonstrated that the developed model maintains consistent predictive performance across different subsets of the dataset, indicating good generalization capability and a low risk of overfitting.

8.8 Feature Importance Analysis

Feature importance analysis was performed to quantify the contribution of each predictor variable toward estimating PM_{2.5} concentrations.

The importance scores generated by the Random Forest model are presented in Table 8.4.

Table 8.4: Feature Importance of the Final Random Forest Model

Rank	Feature	Importance
1	QV2M	0.283
2	AOD ₅₅₀	0.169
3	PM ₁₀	0.149
4	T2M	0.096
5	PBLTOP	0.054
6	PS	0.041
7	Year	0.039
8	CO	0.039
9	NO	0.029
10	Wind Speed	0.017
11	NO ₂	0.015
12	Month	0.014
13	SO ₂	0.013
14	City_Encoded	0.013
15	O ₃	0.011
16	NH ₃	0.011
17	NO _x	0.009

Figure 8.1 illustrates the feature importance obtained from the final Random Forest model.

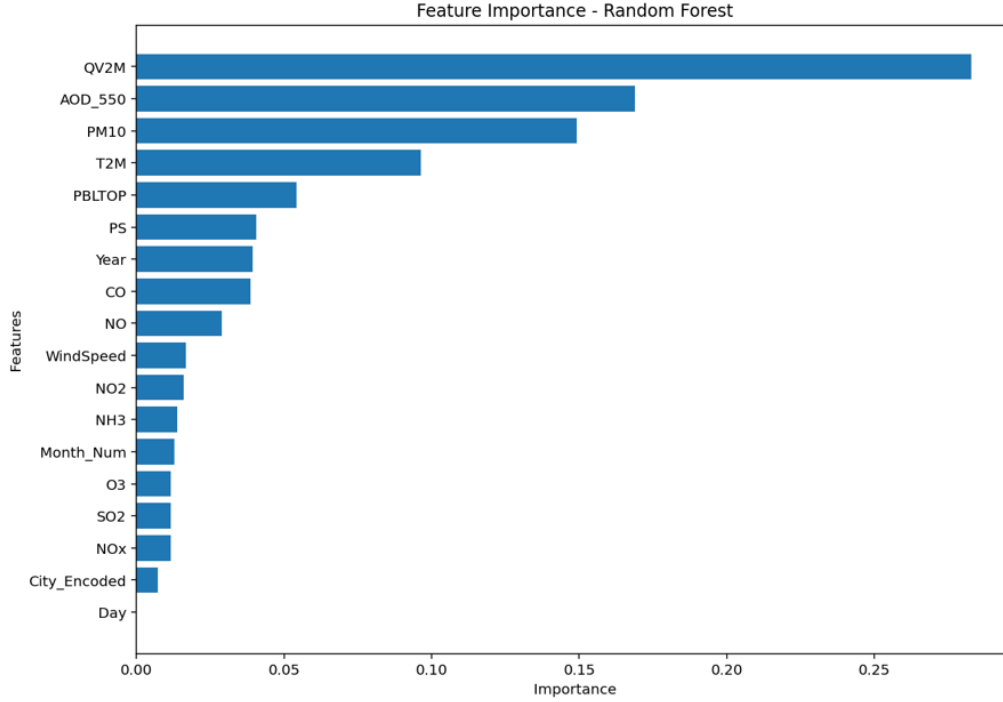


Figure 8.1: Feature Importance of the Final Random Forest Model

The analysis indicates that Specific Humidity (QV2M) is the most influential predictor of $\text{PM}_{2.5}$ concentration, followed by Aerosol Optical Depth (AOD_{550}) and PM_{10} .

The prominence of AOD_{550} demonstrates the importance of incorporating satellite-derived aerosol information, while the high importance of meteorological variables confirms that atmospheric conditions play a significant role in determining particulate matter concentration.

8.9 Comparison Between the Initial and Final Models

The performance of the final Random Forest model was compared with that of the initial model to evaluate the impact of integrating satellite-derived and meteorological information.

Table 8.5 summarizes the comparison.

Table 8.5: Comparison of Initial and Final Random Forest Models

Metric	Initial Model	Final Model
MAE	13.86	13.32
RMSE	31.89	20.53
R^2	0.763	0.820

The incorporation of MODIS MAIAC Aerosol Optical Depth and NASA MERRA-2 meteorological variables significantly improved the predictive capability of the Random Forest model.

The increase in the coefficient of determination from 0.763 to 0.820 demonstrates the effectiveness of multi-source environmental data fusion for $\text{PM}_{2.5}$ estimation. Furthermore, the reductions in MAE and RMSE indicate that the final model produces more accurate and reliable predictions than the initial model.

The results confirm that combining ground-based observations, satellite remote sensing data, and meteorological information provides a comprehensive framework for air quality prediction and substantially enhances the performance of machine learning models.

9. Model Deployment and Spatial Prediction

9.1 Introduction

Following the successful development and evaluation of the final Random Forest model, the trained model was deployed to generate $\text{PM}_{2.5}$ predictions for the complete integrated dataset. Model deployment enables the developed machine learning framework to estimate $\text{PM}_{2.5}$ concentrations without requiring retraining, thereby improving computational efficiency and facilitating future real-world applications.

The deployment stage involved saving the trained Random Forest model, generating predictions for all observations in the integrated dataset, integrating geographic coordinates, and visualizing the predicted air quality over India through spatial mapping.

9.2 Saving the Trained Random Forest Model

After successful training, the Random Forest Regression model was saved using the `Joblib` library. Saving the trained model preserves all learned decision trees and model parameters, allowing the model to be reused for future predictions without repeating the training process.

The trained model was stored as:

`AQIVision_RF_Model.pkl`

The saved model serves as the final inference model for estimating $\text{PM}_{2.5}$ concentrations on unseen datasets.

9.3 Generation of $\text{PM}_{2.5}$ Predictions

The saved Random Forest model was loaded and applied to the final integrated AQIVision dataset to generate predicted $\text{PM}_{2.5}$ concentrations.

Before prediction, the same preprocessing operations used during model training were repeated to maintain consistency between the training and deployment stages. These operations included temporal feature generation, city label encoding, and computation of the Wind Speed feature from the horizontal wind components.

The trained model generated predicted $PM_{2.5}$ values for every observation in the dataset. The predicted values were stored in a new attribute named `Predicted_PM25`, producing a complete prediction dataset for further analysis and visualization.

9.4 Integration of Geographic Coordinates

To enable geographical visualization of the predicted $PM_{2.5}$ concentrations, latitude and longitude information was merged with the prediction dataset.

City coordinates were obtained from the prepared AQIVision coordinate database and integrated using the common City attribute through a left join operation. The resulting dataset contains pollutant measurements, meteorological variables, satellite-derived Aerosol Optical Depth, predicted $PM_{2.5}$ concentrations, and the corresponding geographic coordinates for every study location.

The inclusion of geographic information enables spatial representation of air quality across India and supports subsequent visualization of regional pollution patterns.

9.5 Spatial Distribution of Predicted $PM_{2.5}$

To visualize the geographical distribution of air pollution across India, the average predicted $PM_{2.5}$ concentration was calculated for each city using the generated prediction dataset. The city-wise predictions were aggregated by computing the mean predicted $PM_{2.5}$ value for every study location.

The aggregated dataset was converted into a GeoDataFrame using the GeoPandas library, where each city was represented by its corresponding latitude and longitude coordinates. The administrative boundary of India was used as the base map to provide geographical context for the visualization.

Each city was represented by a circular marker whose colour corresponds to the average predicted $PM_{2.5}$ concentration. A continuous colour scale ranging from green to red was adopted, where green indicates relatively lower pollution levels and red represents higher $PM_{2.5}$ concentrations. This colour scheme enables rapid identification of pollution hotspots and regional variations in air quality.

The generated spatial map provides an intuitive visualization of the model predictions and demonstrates the capability of integrating machine learning with geospatial analysis for environmental monitoring.

Figure 9.1 illustrates the spatial distribution of the average predicted $\text{PM}_{2.5}$ concentrations across the selected Indian cities.

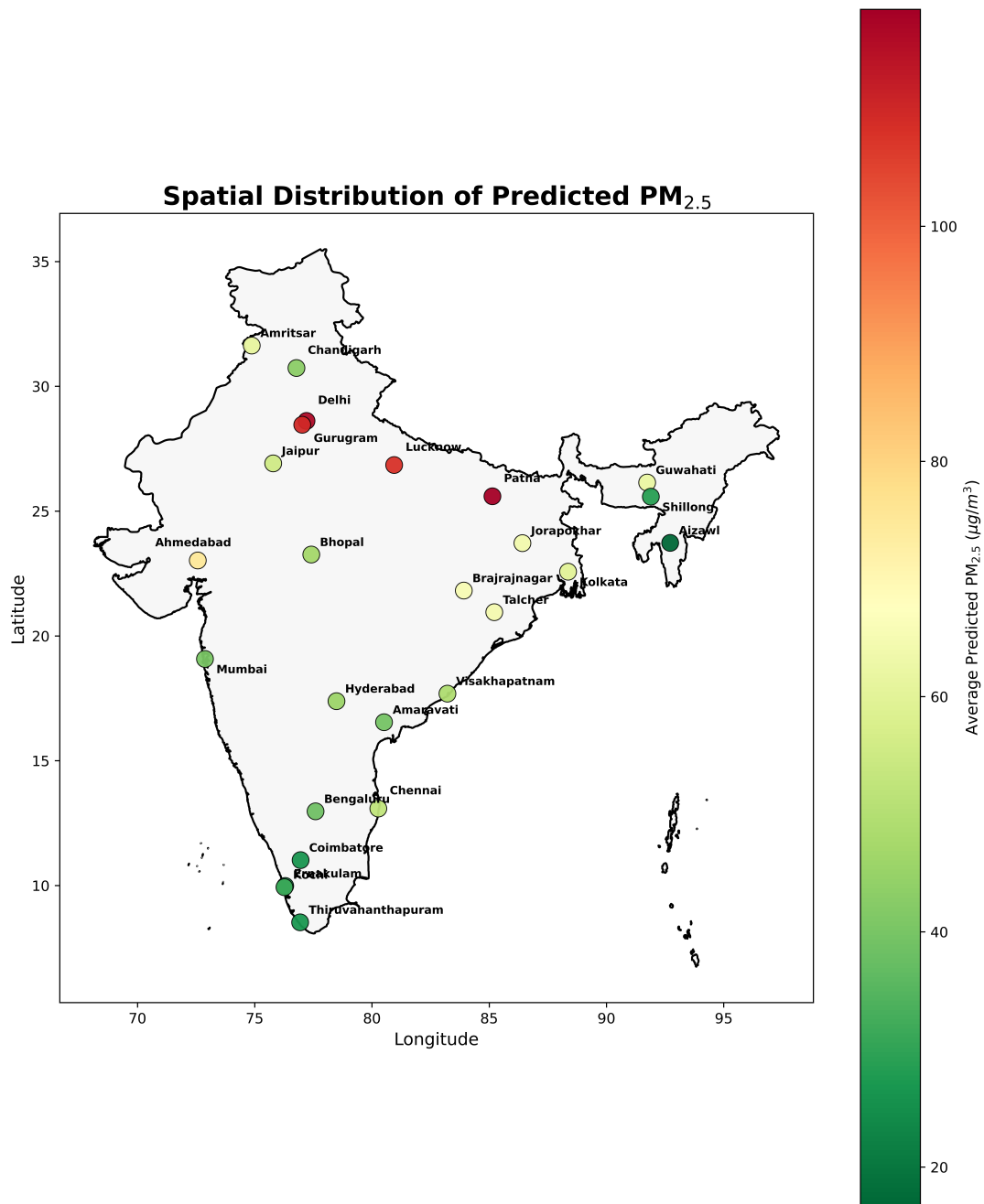


Figure 9.1: Spatial Distribution of Average Predicted $\text{PM}_{2.5}$ Concentrations Across Indian Cities

The spatial distribution reveals noticeable regional differences in air quality across India. Northern cities, particularly Delhi, Lucknow, and Patna, exhibit comparatively higher predicted $\text{PM}_{2.5}$ concentrations, indicating severe particulate pollution. Moderate pollution levels are observed across central and eastern India, whereas several southern cities, including Kochi, Thiruvananthapuram, Coimbatore, and Bengaluru, display comparatively lower predicted $\text{PM}_{2.5}$ concentrations.

The observed spatial pattern is consistent with known regional variations in air pollution and demonstrates that the developed Random Forest model effectively captures geographical trends in PM_{2.5} concentrations. Such spatial visualization provides valuable insights for environmental monitoring, urban planning, and evidence-based policy formulation aimed at improving air quality.

9.6 Conclusion

This project presented the development of **AQIVision**, a machine learning framework for estimating monthly surface-level PM_{2.5} concentrations by integrating ground-based air quality observations, satellite-derived Aerosol Optical Depth (AOD), and meteorological variables. The study successfully demonstrated the effectiveness of multi-source environmental data fusion in improving the prediction of particulate matter concentrations across major Indian cities.

The project involved comprehensive data collection from the Central Pollution Control Board (CPCB), extraction of MODIS MAIAC Aerosol Optical Depth and NASA MERRA-2 meteorological parameters using Google Earth Engine, data preprocessing, feature engineering, and the development of a Random Forest Regression model. The final model achieved an R^2 score of **0.82**, with an MAE of **13.32** and an RMSE of **20.53**, demonstrating a significant improvement over the initial model that relied solely on ground monitoring data.

In addition to model development, the trained Random Forest model was successfully deployed to generate PM_{2.5} predictions for the complete integrated dataset. Geographic coordinates were incorporated into the prediction dataset to facilitate spatial visualization, and a city-wise prediction map was developed to illustrate the spatial distribution of air pollution across India. The generated maps effectively identified regional pollution hotspots and demonstrated the capability of combining machine learning with geospatial analysis for environmental monitoring.

Overall, the results confirm that integrating satellite observations and meteorological information with conventional ground-based measurements substantially enhances PM_{2.5} prediction accuracy. The developed framework provides a scalable and reliable approach for air quality estimation and can serve as a valuable decision-support tool for environmental monitoring agencies, researchers, and policymakers working towards improved air quality management.

Future work may focus on incorporating additional remote sensing datasets, evaluating advanced machine learning and deep learning algorithms, developing real-time prediction systems, and deploying the model through web-based applications to enable continuous monitoring of air quality.